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/ [Test 4 \(topics 10-12: Applications of Lagrange's theorem,Coding theory,Coding theory and group codes\)](#)

Started on	Wednesday, 27 November 2024, 2:17 PM
State	Finished
Completed on	Wednesday, 27 November 2024, 2:37 PM
Time taken	19 mins 29 secs
Marks	2.50/3.00
Grade	2.50 out of 3.00 (83.34%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Partially correct

Mark 0.50 out of 1.00

Bob is using the RSA encryption scheme with public key equal to $(n, x) = (2501, 2023)$.

a) Bob's RSA key is weak. Break his encryption and find his Bob's private key:

$y =$

Your last answer was interpreted as follows:

1687

It might help to know that the following is a list of the first prime numbers:

[2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79]

b) Alice uses Bob's public key to encrypt a message, which she then sends over a public channel.

The encrypted message is

$M = 633$

Find the unencrypted (plain text) message

$m =$

Your last answer was interpreted as follows:

444

The following calculator can be used for computing exponents $\text{mod } (n)$:

Base:
Exponent:
Modulus:
Result:

Question **2**

Correct

Mark 1.00 out of 1.00

Given a noisy binary symmetric channel. The probability of making an error when transmitting a single bit is $p = 0.149$.

The bitstring $w = \mathbf{1111001}$ is transmitted.

In all of the answers below, all answers must be rounded off to the three decimals.

a) What is the probability of making no errors transmitting w ?

Your last answer was interpreted as follows:

0.323

b) What is the probability of making at most 1 bit error transmitting w ?

Your last answer was interpreted as follows:

0.719

b) What is the probability of transmitting $\mathbf{1111001}$ and receiving $r = \mathbf{1111010}$?

Your last answer was interpreted as follows:

0.0099

Question 3

Correct

Mark 1.00 out of 1.00

Let

$$E : (\mathbb{Z}/2)^{\times 3} \rightarrow (\mathbb{Z}/2)^{\times 5}$$

be the encoding function $E(w) = w \cdot G$, given by the generator matrix

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix}.$$

a) Compute the minimum distance between any two codewords which are not equal.

$l_{\min} =$

Your last answer was interpreted as follows:

2

b) Find a **non-zero** codeword c for the encoding E .

$c =$

Your last answer was interpreted as follows:

10010

Any codeword will do, as long as it is not the bitstring consisting only of zeros **000...0**.

c) Find a bitstring x in $(\mathbb{Z}/2)^{\times 5}$ which is **not** a codeword for the encoding E .

$x =$

Your last answer was interpreted as follows:

00001

Syntax guide: Give your answers to **b)** and **c)** as bitstrings. E.g. if you think the answer to a question is "1011", then you should enter **1011**.

Question 4

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 496

Signing code:

Your last answer was interpreted as follows:

30636

◀ [Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)

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